Indian Institute of Science
Department of Electrical Communication Engineering

E9 241 **Digital Image Processing** Aug 2026

Lab Assignment #1

(Due on Monday 31/08/2026 by 10 pm)

Instructions

- For all the questions, write your own functions except when allowed to use in-built functions. Then, provide the wrapper/demo code to run all your functions and obtain results. Your code should be self-contained i.e., one should be able to run your code as is without any modifications.
- Please comment on the code to help the TAs understand your thought process
- For Python, if you use any libraries other than numpy, scipy, scikit-image, opencv, pillow, mat-plotlib, pandas, and default modules, please specify the library that needs to be installed to run your code.
- Along with your code, also submit a PDF with all the results (images or numbers) and inferences
- Put all your files (code files and a report PDF) into a single zip file and submit the zip file. Name the zip file with your name.

Problem # 1 Color Space (10pts): Consider the image 'fabric.png', which is in RGB format, quantized to 8 bits.
Note: For questions (a) and (b), feel free to use built-in functions

a) Convert it to YCbCr format

b) For YCbCr image, compute per channel variance. Note your observation/s.

c) *Chroma Sub-sampling*: Consider Cb and Cr channels. For each 2x2 block in Cb and Cr channel, obtain new image by replacing all elements in the block with the average of the 2x2 block.

d) Where do you expect the chroma sub-sampling to create artifacts? Please explain your intuition.

Problem # 2 Histogram Computation (10pts): Consider the image 'coins.png', which is a gray scale image quantized to 8 bits.

a) Compute the histogram of the image 'coins.png', by finding the frequency of pixels for each intensity level. Plot the histogram. Note your observation.

b) Find the expectation and variance of the intensity using the histogram. Verify the result with the actual values.

c) You decide to quantize the intensities to 6 bits. Compute and plot the histogram of the new image. Is it necessary to take the image, quantize the pixels to 6 bits and then compute the histogram?

Problem # 3 Ostu's Binarization (15pts): Consider again the image 'coins.png':

a) Binarize the image 'coins.png' by finding the optimal threshold th^* by minimizing within class variance $\sigma^2_{Intra}(th)$.

b) Flip each pixel value say p_i to $255 - p_i$ to obtain a new image. What would be the new threshold from Ostu's binarization? Verify.

b) Now, instead of flipping, add a constant positive offset 20 to every pixel of the image (clip the values to max value of 255 if it exceeds 255). Binarize this new image by finding the new threshold.

   Analyze the results and describe how the modification of the pixel value altered the threshold.

Problem # 4 Connected Components: (15 pts): Binarize the image 'quote.png' and use connected component analysis to segment the binary image. Identify the largest connected component corresponding to a character, color it red, and return a modified image where this component is highlighted in red over the original image. Use 8-connectedness for neighborhood of a pixel.